SQL STATEMENT
Like,Wildcard,In and Between

SQL LIKE

The SQL LIKE Operator

* The LIKE operator is used in a WHERE clause to search for a specified pattern in a column.
* There are two wildcards often used in conjunction with the LIKE operator:
* The percent sign (%) represents zero, one, or multiple characters
* The underscore sign (_) represents one, single character

SQL LIKE

Here are some examples showing different LIKE operators with '%' and '_' wildcards:

LIKE Operator	Description
WHERE CustomerName LIKE 'a%'	Finds any values that start with "a"
WHERE CustomerName LIKE '%a'	Finds any values that end with "a"
WHERE CustomerName LIKE '%or%'	Finds any values that have "or" in any position
WHERE CustomerName LIKE '_r%'	Finds any values that have "r" in the second position
WHERE CustomerName LIKE 'a_%'	Finds any values that start with "a" and are at least 2 characters in length
WHERE CustomerName LIKE 'a__%'	Finds any values that start with "a" and are at least 3 characters in length
WHERE ContactName LIKE 'a%o'	Finds any values that start with "a" and ends with "o"

Syntax:
SELECT *column1, column2, ...*
FROM *table_name*
WHERE *columnN* LIKE *pattern*;

SQL LIKE EXAMPLE

The following SQL statement selects all customers with a CustomerName starting with "a":

SELECT * FROM Customers WHERE CustomerName LIKE 'a%';

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Alfred Schmidt	Obere Str. 57	Frankfurt	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK

SQL LIKE EXAMPLE

The following SQL statement selects all customers with a CustomerName ending with "a":

SELECT * FROM Customers WHERE CustomerName LIKE '%a';

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
13	Centro comercial Moctezuma	Francisco Chang	Sierras de Granada 9993	México D.F.	05022	Mexico
30	Godos Cocina Típica	José Pedro Freyre	C/ Romero, 33	Sevilla	41101	Spain
61	Que Delícia	Bernardo Batista	Rua da Panificadora, 12	Rio de Janeiro	02389-673	Brazil

SQL LIKE EXAMPLE

The following SQL statement selects all customers with a CustomerName that have "or" in any position:

SELECT * FROM Customers WHERE CustomerName LIKE '%or%'

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
36	Hungry Coyote Import Store	Yoshi Latimer	City Center Plaza 516 Main St.	Elgin	97827	USA
40	La corne d'abondance	Daniel Tonini	67, avenue de l'Europe	Versailles	78000	France

SQL LIKE EXAMPLE

The following SQL statement selects all customers with a CustomerName that have "r" in the second position:

SELECT * FROM Customers WHERE CustomerName LIKE '_r%';

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
17	Drachenblut Delikatessend	Sven Ottlieb	Walserweg 21	Aachen	52066	Germany
20	Ernst Handel	Roland Mendel	Kirchgasse 6	Graz	8010	Austria
25	Frankenversand	Peter Franken	Berliner Platz 43	München	80805	Germany

SQL LIKE EXAMPLE

The following SQL statement selects all customers with a CustomerName that starts with "a" and are at least 3 characters in length

SELECT * FROM Customers WHERE CustomerName LIKE 'a__%';

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Alfred Schmidt	Obere Str. 57	Frankfurt	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK

SQL LIKE EXAMPLE

The following SQL statement selects all customers with a ContactName that starts with "a" and ends with "o":

SELECT * FROM Customers WHERE ContactName LIKE 'a%o';

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
69	Romero y tomillo	Alejandra Camino	Gran Vía, 1	Madrid	28001	Spain

SQL LIKE EXAMPLE

The following SQL statement selects all customers with a CustomerName that does NOT start with "a":

SELECT * FROM Customers WHERE CustomerName NOT LIKE 'a%';

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden
12	Cactus Comidas para llevar	Patricio Simpson	Cerrito 333	Buenos Aires	1010	Argentina
17	Drachenblut Delikatessend	Sven Ottlieb	Walserweg 21	Aachen	52066	Germany
19	Eastern Connection	Ann Devon	35 King George	London	WX3 6FW	UK
21	Familia Arquibaldo	Aria Cruz	Rua Orós, 92	São Paulo	05442-030	Brazil

SQL Wildcards

* A wildcard character is used to substitute one or more characters in a string.
* Wildcard characters are used with the LIKE operator. The LIKE operator is used in a WHERE clause to search for a specified pattern in a column.

SQL Wildcards Example

The following SQL statement selects all customers with a City starting with "ber":

SELECT * FROM Customers WHERE City LIKE 'ber%';

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
14	Chop-suey Chinese	Yang Wang	Hauptstr. 29	Bern	3012	Switzerland
49	Magazzini Alimentari Riuniti	Giovanni Rovelli	Via Ludovico il Moro 22	Bergamo	24100	Italy

SQL Wildcards Example

The following SQL statement selects all customers with a City containing the pattern "es":

SELECT * FROM Customers WHERE City LIKE '%es%';

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
12	Cactus Comidas para llevar	Patricio Simpson	Cerrito 333	Buenos Aires	1010	Argentina
18	Du monde entier	Janine Labrune	67, rue des Cinquante Otages	Nantes	44000	France
26	France restauration	Carine Schmitt	54, rue Royale	Nantes	44000	France
38	Island Trading	Helen Bennett	Garden House Crowther Way	Cowes	PO31 7PJ	UK
40	La corne d'abondance	Daniel Tonini	67, avenue de l'Europe	Versailles	78000	France
50	Maison Dewey	Catherine Dewey	Rue Joseph-Bens 532	Bruxelles	B-1180	Belgium
54	Océano Atlántico Ltda.	Yvonne Moncada	Ing. Gustavo Moncada 8585 Piso 20-A	Buenos Aires	1010	Argentina
64	Rancho grande	Sergio Gutiérrez	Av. del Libertador 900	Buenos Aires	1010	Argentina
88	Wellington Importadora	Paula Parente	Rua do Mercado, 12	Resende	08737-363	Brazil

SQL Wildcards Example

The following SQL statement selects all customers with a City starting with any character, followed by "ondon":

```
SELECT * FROM Customers WHERE City LIKE '_ondon';
```

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
11	B's Beverages	Victoria Ashworth	Fauntleroy Circus	London	EC2 5NT	UK
16	Consolidated Holdings	Elizabeth Brown	Berkeley Gardens 12 Brewery	London	WX1 6LT	UK
19	Eastern Connection	Ann Devon	35 King George	London	WX3 6FW	UK
53	North/South	Simon Crowther	South House 300 Queensbridge	London	SW7 1RZ	UK
72	Seven Seas Imports	Hari Kumar	90 Wadhurst Rd.	London	OX15 4NB	UK

SQL Wildcards Example

The following SQL statement selects all customers with a City starting with "L", followed by any character, followed by "n", followed by any character, followed by "on":

SELECT * FROM Customers WHERE City LIKE 'L_n_on';

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
11	B's Beverages	Victoria Ashworth	Fauntleroy Circus	London	EC2 5NT	UK
16	Consolidated Holdings	Elizabeth Brown	Berkeley Gardens 12 Brewery	London	WX1 6LT	UK
19	Eastern Connection	Ann Devon	35 King George	London	WX3 6FW	UK
53	North/South	Simon Crowther	South House 300 Queensbridge	London	SW7 1RZ	UK
72	Seven Seas Imports	Hari Kumar	90 Wadhurst Rd.	London	OX15 4NB	UK

SQL Wildcards Example

The following SQL statement selects all customers with a City starting with "b", "s", or "p":

SELECT * FROM Customers WHERE City LIKE '[bsp]%';

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
7	Blondel père et fils	Frédérique Citeaux	24, place Kléber	Strasbourg	67000	France
14	Chop-suey Chinese	Yang Wang	Hauptstr. 29	Bern	3012	Switzerland
48	Lonesome Pine Restaurant	Fran Wilson	89 Chiaroscuro Rd.	Portland	97219	USA

SQL Wildcards Example

The following SQL statement selects all customers with a City starting with "a", "b", or "c":

SELECT * FROM Customers WHERE City LIKE '[a-c]%';

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
12	Cactus Comidas para llevar	Patricio Simpson	Cerrito 333	Buenos Aires	1010	Argentina
14	Chop-suey Chinese	Yang Wang	Hauptstr. 29	Bern	3012	Switzerland
17	Drachenblut Delikatessend	Sven Ottlieb	Walserweg 21	Aachen	52066	Germany
24	Folk och fä HB	Maria Larsson	Åkergatan 24	Bräcke	S-844 67	Sweden
29	Galería del gastrónomo	Eduardo Saavedra	Rambla de Cataluña, 23	Barcelona	08022	Spain
31	Gourmet Lanchonetes	André Fonseca	Av. Brasil, 442	Campinas	04876-786	Brazil
33	GROSELLA-Restaurante	Manuel Pereira	5ª Ave. Los Palos Grandes	Caracas	1081	Venezuela

SQL Wildcards Example

The two following SQL statements select all customers with a City NOT starting with "b", "s", or "p":

```
SELECT * FROM Customers WHERE City LIKE '[!bsp]%';
```

OR

```
SELECT * FROM Customers WHERE City NOT LIKE '[bsp]%';
```

SQL IN Operator

The SQL IN Operator

* The IN operator allows you to specify multiple values in a WHERE clause.
* The IN operator is a shorthand for multiple OR conditions.

IN Operator Examples

The following SQL statement selects all customers that are located in "Germany", "France" or "UK":

SELECT * FROM Customers WHERE Country IN ('Germany', 'France', 'UK');

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Alfred Schmidt	Obere Str. 57	Frankfurt	12209	Germany
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
6	Blauer See Delikatessen	Hanna Moos	Forsterstr. 57	Mannheim	68306	Germany
7	Blondel père et fils	Frédérique Citeaux	24, place Kléber	Strasbourg	67000	France
9	Bon app'	Laurence Lebihans	12, rue des Bouchers	Marseille	13008	France
11	B's Beverages	Victoria Ashworth	Fauntleroy Circus	London	EC2 5NT	UK
16	Consolidated Holdings	Elizabeth Brown	Berkeley Gardens 12 Brewery	London	WX1 6LT	UK
17	Drachenblut Delikatessend	Sven Ottlieb	Walserweg 21	Aachen	52066	Germany
18	Du monde entier	Janine Labrune	67, rue des Cinquante Otages	Nantes	44000	France
19	Eastern Connection	Ann Devon	35 King George	London	WX3 6FW	UK
23	Folies gourmandes	Martine Rancé	184, chaussée de Tournai	Lille	59000	France
25	Frankenversand	Peter Franken	Berliner Platz 43	München	80805	Germany
26	France restauration	Carine Schmitt	54, rue Royale	Nantes	44000	France
38	Island Trading	Helen Bennett	Garden House Crowther Way	Cowes	PO31 7PJ	UK
39	Königlich Essen	Philip Cramer	Maubelstr. 90	Brandenburg	14776	Germany
40	La corne d'abondance	Daniel Tonini	67, avenue de l'Europe	Versailles	78000	France

IN Operator Examples

The following SQL statement selects all customers that are NOT located in "Germany", "France" or "UK":

SELECT * FROM Customers WHERE Country NOT IN ('Germany', 'France', 'UK');

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden
8	Bólido Comidas preparadas	Martín Sommer	C/ Araquil, 67	Madrid	28023	Spain
10	Bottom-Dollar Marketse	Elizabeth Lincoln	23 Tsawassen Blvd.	Tsawassen	T2F 8M4	Canada
12	Cactus Comidas para llevar	Patricio Simpson	Cerrito 333	Buenos Aires	1010	Argentina
13	Centro comercial Moctezuma	Francisco Chang	Sierras de Granada 9993	México D.F.	05022	Mexico
14	Chop-suey Chinese	Yang Wang	Hauptstr. 29	Bern	3012	Switzerland
15	Comércio Mineiro	Pedro Afonso	Av. dos Lusíadas, 23	São Paulo	05432-043	Brazil
20	Ernst Handel	Roland Mendel	Kirchgasse 6	Graz	8010	Austria
21	Familia Arquibaldo	Aria Cruz	Rua Orós, 92	São Paulo	05442-030	Brazil
22	FISSA Fabrica Inter. Salchichas S.A.	Diego Roel	C/ Moralzarzal, 86	Madrid	28034	Spain
24	Folk och fä HB	Maria Larsson	Åkergatan 24	Bräcke	S-844 67	Sweden
27	Franchi S.p.A.	Paolo Accorti	Via Monte Bianco 34	Torino	10100	Italy
28	Furia Bacalhau e Frutos do Mar	Lino Rodriguez	Jardim das rosas n. 32	Lisboa	1675	Portugal

IN Operator Examples

The following SQL statement selects all customers that are from the same countries as the suppliers

SELECT * FROM Customers WHERE Country IN (SELECT Country FROM Suppliers);

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Alfred Schmidt	Obere Str. 57	Frankfurt	12209	Germany
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden
6	Blauer See Delikatessen	Hanna Moos	Forsterstr. 57	Mannheim	68306	Germany
7	Blondel père et fils	Frédérique Citeaux	24, place Kléber	Strasbourg	67000	France
8	Bólido Comidas preparadas	Martín Sommer	C/ Araquil, 67	Madrid	28023	Spain
9	Bon app'	Laurence Lebihans	12, rue des Bouchers	Marseille	13008	France
10	Bottom-Dollar Marketse	Elizabeth Lincoln	23 Tsawassen Blvd.	Tsawassen	T2F 8M4	Canada
11	B's Beverages	Victoria Ashworth	Fauntleroy Circus	London	EC2 5NT	UK
15	Comércio Mineiro	Pedro Afonso	Av. dos Lusíadas, 23	São Paulo	05432-043	Brazil
16	Consolidated Holdings	Elizabeth Brown	Berkeley Gardens 12 Brewery	London	WX1 6LT	UK
17	Drachenblut Delikatessend	Sven Ottlieb	Walserweg 21	Aachen	52066	Germany

SQL BETWEEN Operator

* The BETWEEN operator selects values within a given range. The values can be numbers, text, or dates.
* The BETWEEN operator is inclusive: begin and end values are included.

SELECT *column_name(s)*
FROM *table_name*
WHERE *column_name* BETWEEN *value1* AND *value2;*

BETWEEN Example

The following SQL statement selects all products with a price between 10 and 20:

SELECT * FROM Products WHERE Price BETWEEN 10 AND 20;

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
1	Chais	1	1	10 boxes x 20 bags	18
2	Chang	1	1	24 - 12 oz bottles	19
3	Aniseed Syrup	1	2	12 - 550 ml bottles	10
15	Genen Shouyu	6	2	24 - 250 ml bottles	15.5
16	Pavlova	7	3	32 - 500 g boxes	17.45
21	Sir Rodney's Scones	8	3	24 pkgs. x 4 pieces	10
25	NuNuCa Nuß-Nougat-Creme	11	3	20 - 450 g glasses	14
31	Gorgonzola Telino	14	4	12 - 100 g pkgs	12.5
34	Sasquatch Ale	16	1	24 - 12 oz bottles	14
35	Steeleye Stout	16	1	24 - 12 oz bottles	18
36	Inlagd Sill	17	8	24 - 250 g jars	19
39	Chartreuse verte	18	1	750 cc per bottle	18

BETWEEN Example

To display the products outside the range of the previous example, use NOT BETWEEN:

```
SELECT * FROM Products WHERE Price NOT BETWEEN 10 AND 20;
```

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22
5	Chef Anton's Gumbo Mix	2	2	36 boxes	21.35
6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25
7	Uncle Bob's Organic Dried Pears	3	7	12 - 1 lb pkgs.	30
8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars	40
9	Mishi Kobe Niku	4	6	18 - 500 g pkgs.	97
10	Ikura	4	8	12 - 200 ml jars	31
11	Queso Cabrales	5	4	1 kg pkg.	21
12	Queso Manchego La Pastora	5	4	10 - 500 g pkgs.	38
13	Konbu	6	8	2 kg box	6

BETWEEN Example

The following SQL statement selects all products with a price between 10 and 20. In addition; do not show products with a CategoryID of 1,2, or 3:

```
SELECT * FROM Products WHERE Price BETWEEN 10 AND 20 AND CategoryID NOT IN (1,2,3);
```

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
31	Gorgonzola Telino	14	4	12 - 100 g pkgs	12.5
36	Inlagd Sill	17	8	24 - 250 g jars	19
40	Boston Crab Meat	19	8	24 - 4 oz tins	18.4
42	Singaporean Hokkien Fried Mee	20	5	32 - 1 kg pkgs.	14
46	Spegesild	21	8	4 - 450 g glasses	12
57	Ravioli Angelo	26	5	24 - 250 g pkgs.	19.5
58	Escargots de Bourgogne	27	8	24 pieces	13.25
73	Röd Kaviar	17	8	24 - 150 g jars	15
74	Longlife Tofu	4	7	5 kg pkg.	10

BETWEEN Example

The following SQL statement selects all products with a ProductName between Carnarvon Tigers and Filo Mix:

SELECT * FROM ProductsWHERE ProductName BETWEEN 'Carnarvon Tigers' AND 'Filo Mix'

ORDER BY ProductName;

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
18	Carnarvon Tigers	7	8	16 kg pkg.	62.5
1	Chais	1	1	10 boxes x 20 bags	18
2	Chang	1	1	24 - 12 oz bottles	19
39	Chartreuse verte	18	1	750 cc per bottle	18
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22
5	Chef Anton's Gumbo Mix	2	2	36 boxes	21.35
48	Chocolade	22	3	10 pkgs.	12.75
38	Côte de Blaye	18	1	12 - 75 cl bottles	263.5
58	Escargots de Bourgogne	27	8	24 pieces	13.25
52	Filo Mix	24	5	16 - 2 kg boxes	7

BETWEEN Example

The following SQL statement selects all products with a ProductName between Carnarvon Tigers and Chef Anton's Cajun Seasoning:

```
SELECT * FROM Products
WHERE ProductName BETWEEN "Carnarvon Tigers" AND "Chef Anton's Cajun Seasoning"
ORDER BY ProductName;
```

Number of Records: 5

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
18	Carnarvon Tigers	7	8	16 kg pkg.	62.5
1	Chais	1	1	10 boxes x 20 bags	18
2	Chang	1	1	24 - 12 oz bottles	19
39	Chartreuse verte	18	1	750 cc per bottle	18
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22

BETWEEN Example

The following SQL statement selects all products with a ProductName not between Carnarvon Tigers and Mozzarella di Giovanni:

```
SELECT * FROM Products
WHERE ProductName NOT BETWEEN 'Carnarvon Tigers' AND 'Mozzarella di Giovanni'
ORDER BY ProductName;
```

BETWEEN Example

The following SQL statement selects all orders with an OrderDate between '01-July-1996' and '31-July-1996':

```
SELECT * FROM Orders
WHERE OrderDate BETWEEN #07/01/1996# AND #07/31/1996#;
```

OR

```
SELECT * FROM Orders
WHERE OrderDate BETWEEN '1996-07-01' AND '1996-07-31';
```